

Failure budgets and service objectives

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Evaluation pipelines

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Testing model-dependent behaviour

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Authentication and authorisation

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Failure budgets and service objectives. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Failure budgets and service objectives. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Objective 95% over 80 eligible requests. Five fail. Compute allowed failures and observed success.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Objective 95% over 80 eligible requests. Five fail. Compute allowed failures and observed success.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled failure budgets and service objectives evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of failure budgets and service objectives. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Failure budgets and service objectives

A reliability allowance. A practice service requires at least 98 successes in each window of 100 eligible requests.

Current window has 97 successes and 3 failures. Success includes a correct refusal within the agreed time limit.

Failure count in one 100-request window

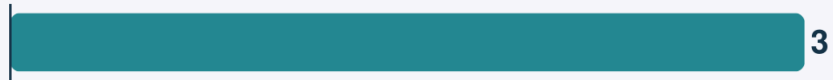
failed requests out of 100

Allowed maximum



2

Observed



3

Bar length shows the supplied value; the marked baseline is zero.

Failure budgets and service objectives

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define success

Use the same predeclared correctness and timing criteria.



2. Compute allowance

The target permits at most 2 failures in 100 requests.



3. Compare record

Three failures misses the target by one request.

CHANGE ONE THING

Repair one failed request into a success in this window.

Explain the changed result using the matching lesson.

Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A reliability allowance. A practice service requires at least 98 successes in each window of 100 eligible requests.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three failures misses the target by one request.

1. Define success
2. Compute allowance
3. Compare record

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare record Three failures misses the target by one request. An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

Practice 2: explain the deciding evidence

Changed-case practice: Repair one failed request into a success in this window. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. 98 successes and 2 failures meet the stated window target.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: 98 successes and 2 failures meet the stated window target. One recorded outcome changes while the denominator stays 100.

Repair a missing building block

Review Evaluation pipelines (ai-6/10). Goal: Explain the mechanism and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the mechanism and terminology of testing model-dependent behaviour. Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the mechanism and terminology of authentication and authorisation. Short practice: Explain why Jo's valid volunteer login does not permit editing C. Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
